Project Title not Assignment Topic here

BMDS2133 Image Processing
Assignment Documentation

Semester 2026xx

Group Members :

(a) Student Name — Algorithm/Module In-Charged

(b) Student Name — Algorithm/Module In-Charged

(c) Student Name — Algorithm/Module In-Charged

(d) Student Name — Algorithm/Module In-Charged

Programme : Bachelor in Data Science (Honours)

Tutor : Your Tutor Name here

Table of Contents

Introduction

1.1 Problem Background

Instruction to Student: Provide a clear narrative introduction to your project and its context. Ensure that every keyword in your project title is defined and explained in its own dedicated paragraph. Do not use bullet points or numbered lists here. All claims, statistics, and background facts must be backed by academic citations in APA format. Write strictly in UK English and avoid first-person pronouns (e.g., do not use "I", "we", "my").

- *Citation needed for the problem that you mentioned*

1.2 Objectives/Aims

Instruction to Student: State <u>maximum of three</u> clear, concise objectives that you aim to achieve. Every objective must begin with the word "To..." and must strictly follow the SMART framework (Specific, Measurable, Achievable, Relevant, and Time-bound). These will be directly evaluated against your final results in Chapters 4 and 5.

- *Understand more from here: [SMART Goals: A How to Guide]*
- *Example:*
    - *Objective 1: To [Action Verb] [Specific, Measurable Outcome]*
    - *Objective 2: To [Action Verb] [Specific, Measurable Outcome]*
    - *Objective 3: To [Action Verb] [Specific, Measurable Outcome]*
- *Make sure all of that can be measurable within experiments*

1.3 Motivation

Instruction to Student: Explain why this work matters. Discuss the potential commercialisation value or the tangible social impact of your project. You must explicitly link your project to broader societal frameworks, such as the ***United Nations Sustainable Development Goals (SDGs)*** *or the* ***Malaysia MADANI*** *concept.*

Figure 1.1 SDG9 Industry Innovation and Infrastructure (United Nations, 2015)

Figure 1.2 Malaysia Madani (Jabatan Penerangan Malaysia, 2023)

Literature Review

2.1 Main Concept / Algorithm 1

Instruction to Student: Provide a deep, well-studied academic overview of the first primary concept, technology, or algorithm used in your project. Support your explanations with credible citations.

2.2 Main Concept / Algorithm 2

Instruction to Student: Provide a deep, well-studied academic overview of the second primary concept, technology, or algorithm used in your project. Support your explanations with credible citations.

2.x Main Concept / Algorithm 3 - *add more if needed*

Instruction to Student: Provide a deep, well-studied academic overview of the third primary concept, technology, or algorithm used in your project. Support your explanations with credible citations.

2.3 Analysis of the existing algorithm (Comparison)

Instruction to Student: Critically analyse and contrast the existing algorithms or systems. You must use the comparison table below to show points of contrast. Feel free to add more columns or rows as necessary to make your analysis comprehensive.

Table 2.1 Comparison of Contemporary Approaches in Image Segmentation

Algorithm / System	**Example of Previous Work *(insert as citation)***	**Advantages**	**Limitations**	**Remarks**
Algorithm 1: [Name]				

Algorithm 2: [Name]				
Algorithm 3: [Name]				

Table 2.2 Comparison of Contemporary Approaches in Contrast Enhancement

Algorithm / System	**Example of Previous Work *(insert as citation)***	**Advantages**	**Limitations**	**Remarks**
Algorithm 1: [Name]				
Algorithm 2: [Name]				
Algorithm 3: [Name]				

Methodology

3.1 System flowchart

Instruction to Student: Insert a high-resolution, logically sound system flowchart detailing the execution and architecture of your system. Reference the diagram in your text (e.g., "As illustrated in Figure 3.1..."). Numbering following the Chapter not section.

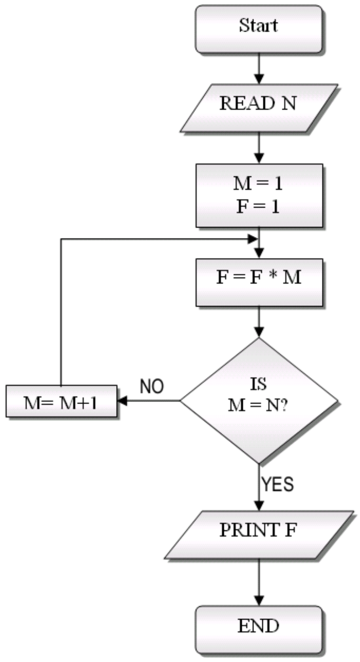

Figure 3.1: System Architecture and Process Flowchart.

3.2 Description of dataset (if Applicable)

Instruction to Student: Clearly state the source of your data (provide the exact URL link if obtained online, or detail the collection method if self-collected). Outline the data structures or provide a comprehensive Data Dictionary. Include a sample image or snippet of the raw dataset.

- *Dataset Source: [Insert Link / Collection Details]*
- *Data Structures / Dictionary: [Define fields, data types, and descriptions]*
- *Sample image*

3.3 Applications of the algorithm(s)

Instruction to Student: Describe exactly how the selected algorithm(s) or technique(s) are practically applied, coded, or integrated within your specific system implementation.

- *Pseudocode with sample input and output*

Algorithm 1 Algorithm 1

Input: (C, S_k)

Output: $\mathrm{HashSession}(1, \underline{r}, C)$ or $\mathrm{HashSession}(0, \rho, C))$

$c \leftarrow \mathrm{Decode}(\underline{c})$

$c.(3f) \in \mathcal{R}/q$

$e \leftarrow (\mathrm{Rounded}(c.(3f)) \bmod 3) \in \mathcal{R}/3$

$e.(1/g) \in \mathcal{R}/3$

$r' \leftarrow \mathrm{Lift}(e.(1/g)) \in \mathcal{R}/q$

$h.r' \in \mathcal{R}/q$

$c' \leftarrow \mathrm{Round}(h.r')$

$\underline{c}' \leftarrow \mathrm{Encode}(c')$

$C' \leftarrow (\underline{c}', \mathrm{HashConfirm}(\underline{r}', \underline{h}))$

if $C' == C$ **then**

  return $\mathrm{HashSession}(1, \underline{r}, C)$

else

  return $\mathrm{HashSession}(0, \rho, C))$

end if

Result & Discussion

4.1 Experimental Results

Instruction to Student: Demonstrate the definitive outcomes of your project based on your test plan. Use suitable data visualisations (e.g., bar charts, stacked line graphs, or pie charts) to illustrate performance or output metrics. Working prototypes as screenshots can be inserted but the results still be the focus.

4.2 Discussion and Critical Interpretation

Instruction to Student: Critically analyse your experimental results. Do not just state what the data is—interpret what the data means. Discuss the implications of your findings and provide a detailed comparison of how different algorithms or settings performed against each other.

Conclusion

5.1 Achievements

Instruction to Student: Synthesise your findings and explicitly state whether your project successfully fulfilled the SMART objectives outlined in Section 1.2. Declare which algorithm proved to be the best based on your experimental data.

- **Objective 1 Status:** [Fulfilled / Partially Fulfilled] – *Briefly explain how based on experimental data.*
- **Objective 2 Status:** [Fulfilled / Partially Fulfilled] – *Briefly explain how based on experimental data.*

- **Objective 3 Status:** [Fulfilled / Partially Fulfilled] – *Briefly explain how based on experimental data.*

Project Code Repository: *[Insert your GitHub Repository Link here, if applicable]*

5.2 Limitations and Future Works

Instruction to Student: Honestly discuss the current constraints, bugs, or boundaries of your prototype. Suggest concrete, actionable improvements or features that could be implemented in future iterations of the work.

References

Instruction to Student: Compile an exhaustive bibliography of all scholarly material referenced within this assignment. You must strictly adhere to the APA referencing format, ensuring each entry includes the author, publication year, title, and publisher or URL. Every in-text citation must have a matching full entry.

- *Specify the Digital Object Identifier (DOI) for your dataset where available.*
- *Avoid websites/forums/researchgate/arxiv if possible.*
- *Numbered lists are not permitted in this section.*
- *Organise all bibliographic entries in alphabetical order.*
- *Employ academic search engines or university library databases to ensure the precision and standardisation of your reference metadata.*

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Appendix A

TUNKU ABDUL RAHMAN UNIVERSITY OF MANAGEMENT AND TECHNOLOGY
Group Contract Form

Faculty	: Faculty of Computing and Information Technology (FOCS)
Programme	:
Year of Study	:
Semester	:
Course Title	: Image Processing
Course Code	: BMDS2133
Group Assignment / Project Title	:
Name of Tutor	:
Tutorial Group	:

Student Name and ID	Role(s)/Task(s) Assigned	Signature and Date

Appendix B

AI Usage Disclosure Form

Student Name : ______________________________

Programme : ______________________________

Course : ______________________________

Assessment Title : ______________________________

1. AI Tool(s) Used (Check that all that apply)

- ☐ None (I did not use any AI tools for this assignment)
- ☐ ChatGPT (Version: __________)
- ☐ Deepseek
- ☐ Gemini
- ☐ Other: ____________________

2. Nature of Assistance (Check all that apply)

- ☐ **Brainstorming:** Generating ideas, topics, or outlines.
- ☐ **Research:** Summarising long articles or finding concepts.
- ☐ **Editing:** Checking grammar, spelling, or sentence structure.
- ☐ **Coding/Math:** Debugging code or explaining a formula.
- ☐ **Creation:** Generating images, data, or draft text.

3. Disclosure Statement

Provide a brief explanation of how AI tools were utilised in completing the task and describe the measures taken to ensure that the final submission reflects your own independent work.

Example:
"I used ChatGPT to structure my initial ideas into an outline. Subsequently, I composed the full paragraphs independently and cross-checked the referenced dates against my textbook to ensure accuracy."

Your Statement:

4. Verification & Integrity

- ☐ I have reviewed and verified the accuracy of all facts, data, and citations generated with the assistance of AI tools.
- ☐ I have properly acknowledged and cited any AI-generated text, ideas, or content in accordance with my instructor's requirements.
- ☐ I confirm that the reasoning, interpretation, and analysis presented in the final submission are entirely my own.

Student Signature : ____

Date :____

AMM 5.3.2026/SMC 1.4.2026